CHARAN SAI REDDY VENKATAPURAM

Mechanical Product Engineer — Hardware from Brief to Field Deployment

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Portfolio: Design Portfolio 2026 (attached)

Graduate Route visa to 27 Feb 2027 • eligible for Skilled Worker sponsorship under the New Entrant provisions

PROFESSIONAL SUMMARY

Mechanical engineer with six years' experience designing physical hardware that has reached real users and survived in the field. I own projects from brief to delivery: material, process and tolerance decisions; drawings and BOMs a supplier can build from without guessing; then the prototype build, the testing and the iteration. I work on the bench as much as on screen — jigs and fixtures built by hand, full-scale models made to settle geometry, a vehicle designed, built and road-tested. 37 units of one design entered public service across 37 districts within a month of release, built on an operation I set up and staffed myself. MSc Engineering Design, Loughborough University.

SKILLS

Mechanical Design & DFM: SolidWorks (CSWA certified, CSWP in progress) — assemblies, mechanisms, sheet metal, welded frames; materials selection, fastening and joining; fabrication and assembly drawings, bills of materials and build guides a supplier can work from; design for the processes and suppliers actually available

Build, Test & Iterate: Prototype and first-article build, assembly, commissioning, debug and rework; jigs, bend templates and timber formers made by hand to agree tooling before cutting metal; concept down-selection on printed parts rather than on screen; full-scale models built to settle geometry; dimensional validation against CAD

Hardware for the Field: Ingress sealing — rubber-beaded perimeters, silicone-sealed joints, sealed cable entries; serviceability designed around the maintaining engineer; key-locked tamper resistance and bonded impact glazing for unattended public equipment; cable routing, clamping and strain relief; motors, printers, displays, cameras, RFID and NFC readers and power supplies integrated into assemblies

Manufacturing Processes: Machining, sheet-metal forming, tube bending, welded fabrication, powder coating and FDM; metals — hot- and cold-rolled steel, IS 2062, AISI 1020, 4130 chromoly, stainless; polymers — PETG, PLA, ABS and TPU, selected on function, with print orientation controlled for finished surfaces; MJF-processed PA evaluated for higher volume

Delivery & Team: Projects owned end to end against fixed tender and production deadlines; supplier selection, qualification and on-site collaboration through build; assembly sequence definition and technician training; static structural FEA in ANSYS and SolidWorks Simulation, checked against analytical models in MATLAB

PROFESSIONAL EXPERIENCE

Mechanical Design Engineer (Freelance) | Sai Rams Innomations Pvt. Ltd.

March 2025 – Present • Remote • Kiosk and access-control hardware manufacturer

- Designed a facial-recognition access terminal through **three complete versions in eight months**, each deployed to customer sites: a deliberately plain first build to prove assembly sequence and serviceability, a second on an internal carrier frame so the shell could change independently, and a third resolved into a single continuous slab with the display flush in a close-tolerance aperture and every fixing, cable and service point moved to the rear.
- Engineered the terminal for unattended sites: a pull-out device tray with cable guide and positive locking, the charger captured inside the wall plate so no cabling is accessible once installed, and a 2 mm acrylic impact window bonded into a seating recess.
- Developing a wall-mounted RFID reader enclosure from five concept directions, down-selected by printing and handling each rather than reviewing on screen; PETG specified across the product family for RF transparency, service temperature and print reliability.
- Planned the manufacturing route against volume: FDM now, with print orientation controlled for a fine textured finish straight off the machine; a hybrid printed front and folded powder-coated aluminium rear defined for the next version; MJF-processed PA under evaluation for outdoor durability.

Mechanical Engineer | Evionics Innovations Pvt. Ltd.

May 2022 – January 2023 • India • Electric two- and three-wheeler startup

- Took an electric cargo two-wheeler from **concept to road-tested prototype**: a bespoke welded roll cage with compound bends in multiple planes on a repurposed donor chassis, integrating a rear hub motor, front and rear disc brakes and a thumb throttle adapted in-house. Designed it with another engineer, built it with the fabricator, commissioned it and rode it.
- Solved a forming problem with **no tooling spend**: when the frame's three-dimensional bends exceeded the fabricator's die-bending capability, decomposed the geometry into numbered 1:1 templates on A1 sheets — bend start, end, sequence and axis per plane — and worked through the build on-site with the fabricator.
- De-risked the material choice before committing: prototyped in IS 2062 tube as an available proxy for AISI 1020 to validate geometry and fit-up, accepting mid-span forming variance where end coordinates met specification rather than thinning the tube. Load paths confirmed by static structural FEA in ANSYS.

Mechanical Design Engineer | Sai Rams Innomations Pvt. Ltd.
September 2018 – May 2022 • India • Kiosk and access-control hardware manufacturer

- Established the company's self-service machine product line from a standing start, with no existing products, documentation or domain experience in-house, and carried it through **three generations to a deployed government product**.
- Won a national tender for the Election Commission of India by packaging a full-size Datacard card printer into roughly a quarter of the competing bid's footprint. Concept to production in two months with an outsourced fabricator; **37 units in service across 37 districts within a month of release**.
- Set up a temporary in-house build operation for that production run: organised the workspace, defined the assembly sequence, and trained two trade-qualified fitters to build and fit the units, supporting them through first-article build and rework.
- Designed the service architecture around the maintaining engineer: a gas-strut hood carrying the touchscreen opens flat to expose both the card printer and the display's rear connections — two service zones from one opening, in a footprint too tight for a swing door.
- Sealed the enclosure for unconditioned deployment sites, and designed 3D-printed housings to locate and shroud the QR/barcode scanner and RFID card reader in the fascia.
- Owned material and supplier decisions: 1.6 mm powder-coated hot-rolled steel, chosen for fabricator stock availability and cost, and the touchscreen vendor sourced and qualified independently. Set touch height and screen rake from the ADA forward-reach limit, validated by building and comparing two full-scale physical models.

Team Design Lead | ISNEE Quad-TORC national ATV competition
October 2016 – October 2017 • India

- Led roll cage design for a national off-road vehicle competition entry, specifying 4130 chromoly for multi-axis off-road loading and running static structural FEA with front and side impact load cases.
- Dimensioned wheelbase and frame envelope to ±0.5 mm in CAD and verified them on the welded structure; machined disc brake flanges, wheel hubs and a rear swing arm in-house and integrated donor driveline, braking and steering components into the bespoke chassis.

PERSONAL PROJECTS

Bottle Armour | removable drop-protection sleeve for a stainless-steel water bottle
Self-initiated • design, prototyping and iteration

- Mapped the bottle's compound curvature with digital calipers and resolved a 0.15 mm interference fit across three printed fit-check iterations, so the sleeve holds under gravity yet stays removable by hand for cleaning.
- Replaced the solid section with a variable-cell lattice in TPU 95A to absorb impact through localised cell buckling, with cell angles held above 45° so the part prints support-free and needs no post-processing.

EDUCATION

MSc Engineering Design | Loughborough University, UK
October 2023 – January 2025

- Thesis: a compliant double parallel guiding flexure investigated as a replacement for the coil spring in a domestic gas-meter safety valve, sized against a 25 mbar over-pressure threshold. Modelled analytically in MATLAB and validated against ANSYS static structural FEA with mesh convergence per load case; agreement within 3%.

B.Tech Mechanical Engineering | Gandhi Institute of Technology and Management, India
July 2014 – September 2018

CERTIFICATIONS & ELIGIBILITY

Certifications: Certified SolidWorks Associate (CSWA – Design); CSWP in progress. Currently building formal competence in GD&T (ISO GPS) and design for injection moulding.

Right to work: Graduate Route visa to 27 February 2027, full and unrestricted; eligible to switch to Skilled Worker under the New Entrant provisions.

Other: Full UK driving licence. Based in Coventry, under an hour from Stirchley by car or train.